A deep neural network estimation of brain age is sensitive to cognitive impairment and decline

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Background

Recent advancements in machine learning have enabled the prediction of brain age from neuroimaging-derived measures, which is more sensitive to cognitive decline and Alzheimer's disease (AD) progression than chronological age. However, prior studies have only incorporated structural and conventional diffusion MRI without accounting for partial volume effects. The goal of the present study is to build 3 densely connected neural network models to predict brain age from T1-weighted MRI, free-water (FW) corrected diffusion MRI (dMRI), and combined FW+T1 features, then to evaluate the association of each predicted brain age with cognition (memory, executive function).

Participant Characteristics & Project Workflow

- Data were drawn from Vanderbilt Memory and Aging Project (VMAP), including only cognitively unimpaired (CU) and mild cognitive impairment (MCI) participants with both neuroimaging and cognitive measures (n=295). MCI participants were age-, sex-, and race-matched with CU controls; there was no difference in age at baseline between diagnostic groups. See **Table 1** for cohort demographic and clinical information.

Table 1 – Demographic and Health Characteristics

Measure	Diagnosis at Baseline		p-value
	CU (n=168)	MCI (n=127)	
Cohort Characteristics			
Total number of visits	568	372	-
Longitudinal follow-up (years)	3.10 (1.44)	2.96 (1.43)	0.230
Demographic and Health Characteristics			
Age at baseline (years)	73.10 (7.16)	73.67 (7.41)	0.504
Sex (% female)	42.26	42.52	1.000
Education (years)	16.39 (2.48)	15.09 (2.75)	<0.001
Race (% non-Hispanic white)	86.90	86.61	1.000
APOE-ε4 (% positive)	29.76	44.09	0.016

- T1-weighted MRI data were preprocessed with multi-atlas segmentation to calculate volumes of 132 ROIs then normalized by total intracranial volume. dMRI data were preprocessed using the *PreQual* pipeline and underwent FW correction to calculate FW and FW-corrected fractional anisotropy (FA_{FWcorr}), which were transformed using *ANTs* and quantified within 48 white matter tracts. T1-weighted MRI and FW-corrected dMRI metrics were harmonized separately using the *Longitudinal ComBat* pipeline.
- Memory and executive function composite scores were obtained for each participant; longitudinal memory and executive function slopes were calculated using a linear mixed effects model.
- A densely connected neural network was built with an input layer, 2 fully-connected dense layers, and an output layer. 3 separate models were created using FW, T1, and combined FW+T1 features to predict brain age. See **Figure 1** for model workflow.
- All models were optimized on mean absolute error using 10-fold cross validation.

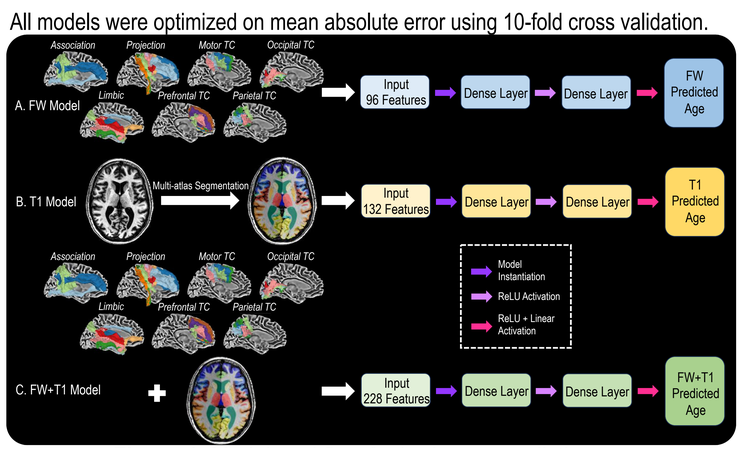

Figure 1. Model workflow for brain age prediction.

- SHAP values were generated for each model to visualize feature importance. Then, logistic regression was conducted with actual age and predicted brain ages as predictor of diagnostic category (CU, MCI); performance was compared using area under the receiver operating characteristic curve (ROC-AUC) and DeLong's test.
- Actual age and predicted brain age (FW, T1, FW+T1) main effect on cross-sectional and longitudinal cognition (memory, executive function) was assessed using linear models; post-hoc competitive model analysis was conducted to determine the unique variance in cross-sectional and longitudinal cognition contributed by each predicted brain age, beyond covariates and actual age. Covariates included diagnosis, sex, race/ethnicity, education, and APOE-ε4 positivity.

Results

Finding 1: All models robustly predicted brain age and demonstrated relevance to aging and AD.

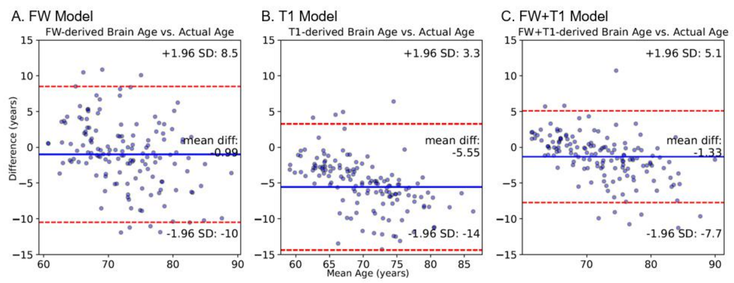

Figure 2. Bland Altman plots for FW-derived age (**A**), T1-derived age (**B**), and FW+T1-derived age (**C**).

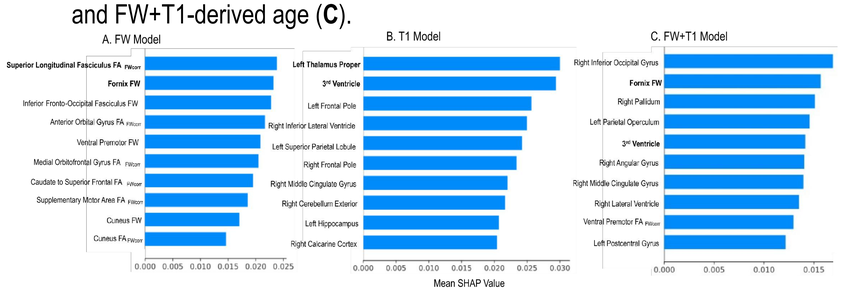

Figure 3. Top 10 most important features for FW-derived (**A**), T1-derived (**B**), and FW+T1-derived (**C**) models. Boldface signifies top features involved in aging and AD.

Finding 2: All predicted brain ages better distinguished diagnostic groups than actual age, though not different from one another in performance.

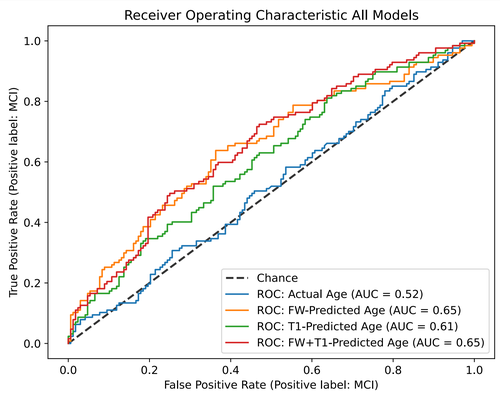

Figure 4. Receiver Operating Characteristic curves for actual, FW-derived, T1-derived, and FW+T1-derived age in predicting diagnostic category (CU, MCI).

Finding 3: Combined FW+T1-derived brain age best predicted cross-sectional and longitudinal memory and executive function.

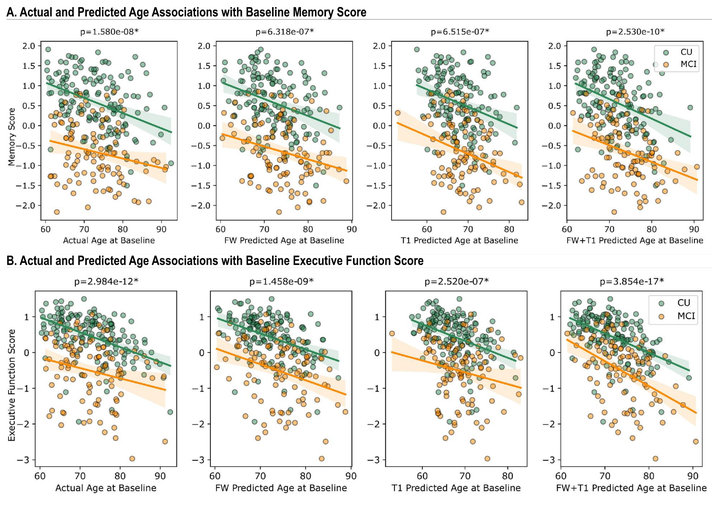

Figure 5. Actual and predicted brain age association with cross-sectional cognition.

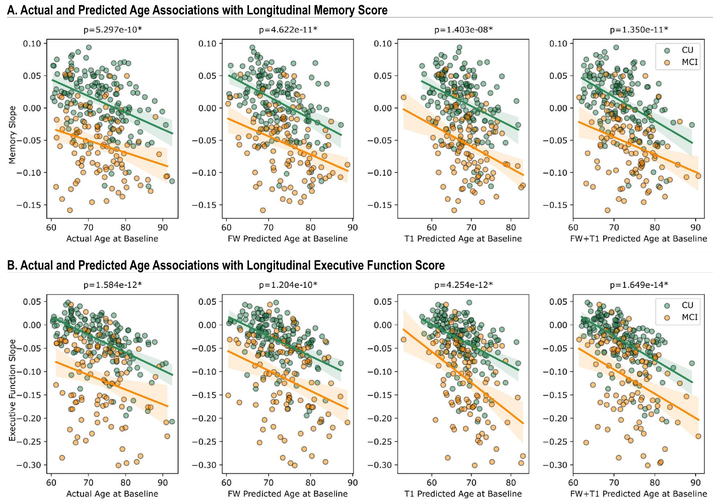

Figure 6. Actual and predicted brain age association with longitudinal cognition.

Conclusions

- Densely connected neural networks robustly predict age, with all models showing top features that are biologically relevant to aging and AD.
- Predicted brain age is a more sensitive measure than actual age to diagnostic category and is associated with memory and executive function impairment and decline; combined FW+T1-derived brain age shows most significant associations.

Findings of this study support neuroimaging-derived brain age as a potential biomarker in assessing AD diagnosis and cognitive decline and suggest that FW correction may improve model performance. Replication of results leveraging multiple cohorts is needed. Future studies should incorporate both gray and white matter features in building machine learning algorithms to study aging and AD.

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